Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
2	(a)		Energy transferred from {some form / chemical} to electrical [potential] (1) {per coulomb / unit charge} [passing through the cell] (1)	2			2		
	(b)	(i)	Use of $R = \frac{V}{I}$ i.e. $\frac{5.4}{0.3}$	1			1	1	
		(ii)	Substitution into $V = E - Ir$ i.e. $5.4 = 6 - 0.3r$ (1) to give $r = 2[\Omega]$ (1) $\div 4 = 0.5[\Omega]$ **ecf** on r (1) **OR** Total resistance of circuit = $\frac{E}{I}$ = 20 [Ω] (1) $20 - 18$ (1) $\div 4 = 0.5$ [Ω] **ecf** on r (1) **OR** Lost volts $= 6 - 5.4 = 0.6$ [V] (1) $\frac{0.6}{0.3} = 2$ [Ω] (1) $\div 4 = 0.5[\Omega]$ **ecf** on r (1)	1	1 1		3	3	
		(iii)	Correct substitution into a power equation using appropriate values e.g. $P = I^2R$ so $0.3^2 \times 2$ (1) = 0.18 [W] (1) **ecf from ii**	1	1		2	2	
		(iv)	$6 = I(9 + 2) \therefore I = 0.55$ [A] (1) $0.55^2 \times 2 = 0.60$ [W] (1) $\frac{0.60}{0.18}$ = 3.3 or 0.60 is more than 2 × 0.18 owtte hence incorrect (1) Accept 0.58 [W] or 0.59 [W] or 0.61 [W]			3	3	3	
			Question 2 total	**5**	**3**	**3**	**11**	**9**	**0**